Raspberry Pi Based Oscilloscope

Terminal Commands and Python Code

Practical: Read analog voltage from an ADS1115 ADC and display it as a live graph.

1. Enable I²C

```
sudo raspi-config
```

Select Interface Options → I²C → Enable, then exit.

2. Update Raspberry Pi

```
sudo apt-get update
sudo apt-get upgrade -y
```

3. Install required system packages

```
sudo apt-get install -y python3-pip python3-dev python3-smbus i2c-tools python3-matplotlib python3-venv
```

4. Create and activate a Python environment

```
python3 -m venv --system-site-packages ~/scope-env
source ~/scope-env/bin/activate
```

5. Install the ADS1115 library

```
pip install adafruit-circuitpython-ads1x15
```

6. Check the I²C device

```
sudo i2cdetect -y 1
```

The ADS1115 commonly appears at address 48; this can vary with its ADDR connection.

Python Code — oscilloscope.py

Open IDLE → File → New File. Paste the code and save it as oscilloscope.py.

```
import time
import board
import busio
import matplotlib.pyplot as plt
from collections import deque
import adafruit_ads1x15.ads1115 as ADS
from adafruit_ads1x15.analog_in import AnalogIn

i2c = busio.I2C(board.SCL, board.SDA)
ads = ADS.ADS1115(i2c)
chan = AnalogIn(ads, ADS.P0)

x_data = deque(maxlen=100)
y_data = deque(maxlen=100)

plt.ion()
fig, ax = plt.subplots()
line, = ax.plot([], [])

ax.set_title("Raspberry Pi Oscilloscope")
ax.set_xlabel("Time (seconds)")
ax.set_ylabel("Voltage (V)")
ax.grid(True)

start = time.time()

try:
    while True:
        x_data.append(time.time() - start)
        y_data.append(chan.voltage)
        line.set_data(x_data, y_data)
        ax.relim()
        ax.autoscale_view()
        plt.pause(0.05)

except KeyboardInterrupt:
    print("Oscilloscope stopped.")

finally:
    plt.close()
```

Run the program

```
source ~/scope-env/bin/activate
python3 ~/oscilloscope.py
```

Connection reminder: ADS1115 VDD → Raspberry Pi 3.3V (physical pin 1 or 17); GND → GND (e.g. physical pin 9); SCL → physical pin 5 (GPIO 3); SDA → physical pin 3 (GPIO 2). Connect the measured signal to A0/P0 and signal ground to GND. Keep the input within the ADC's permitted range.

This is a standalone graph script, not the original graph file downloaded by the manual's Git command.